

Practical-Week1

1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

```
manikanta@Hemu:~$ nano process.c
manikanta@Hemu:~$ gcc process.c -o process
manikanta@Hemu:~$ ./process
Enter a Linux command: ls
Parent Process PID: 627
Child Process PID: 628
Executing command: ls
OSSP      assets          hello      process.c
OS_Lab    command_executor.c      process    shell_project
Parent process: Child execution completed.
```

```
manikanta@Hemu:~$ ./process
Enter a Linux command: date
Parent Process PID: 675
Parent Process PID: 678
Child Process PID: 678
Fri Aug 14 09:53:31 UTC 2026
Parent process: Child execution completed.
manikanta@Hemu:~$ |
```

Report: I ran process.c with two commands: ls (parent PID 627, child PID 628) and date (parent PID 675, child PID 678). Both executed correctly with distinct, valid PIDs, and the parent reported completion after each. This confirms the program correctly forks and execs arbitrary user-specified commands.

2. Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
ajithmutyala@Ajith:~$ uname -a
Linux Ajith 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun
18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
ajithmutyala@Ajith:~$ |
```

```
ajithmutyala@Ajith:~$ lsblk
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/

ajithmutyala@Ajith:~$ |
```

```
ajithmutyala@Ajith:~$ ps
PID TTY          TIME CMD
 936 pts/0        00:00:00 bash
 964 pts/0        00:00:00 ps
ajithmutyala@Ajith:~$ |
```

Report: I checked the system using `uname -a`, `lsblk`, and `ps`. `uname -a` confirmed the kernel (WSL2 Linux). `lsblk` listed disks including root (sdd, 1T) and swap (sdc, 2G). `ps` showed the active session's processes: bash (PID 403) and ps (PID 754). This confirms the environment inspection commands correctly reflect the running system.

ajithmutyala@Ajith:~\$ top

top - 07:45:27 up 2 min, 1 user, load average: 0.97, 0.36, 0.13

Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.1 us, 0.2 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 7753.6 total, 6053.4 free, 550.2 used, 1302.5 buff/cache

MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7203.4 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
530	root	20	0	1681604	12424	10440	S	1.3	0.2	0:00.04	wsl-pro-service
1	root	20	0	21532	13060	9732	S	0.3	0.2	0:00.46	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	42236	16016	14856	S	0.0	0.2	0:00.12	systemd-journal
97	root	20	0	25404	6816	5180	S	0.0	0.1	0:00.09	systemd-udev
174	systemd+	20	0	21460	13212	10956	S	0.0	0.2	0:00.05	systemd-resolve
175	systemd+	20	0	91028	7988	7028	S	0.0	0.1	0:00.03	systemd-timesyn
181	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.00	cron
182	message+	20	0	9540	5420	4728	S	0.0	0.1	0:00.04	dbus-daemon
189	root	20	0	17972	8776	7732	S	0.0	0.1	0:00.04	systemd-logind
205	syslog	20	0	222516	5668	4640	S	0.0	0.1	0:00.04	rsyslogd
207	root	20	0	3124	1976	1836	S	0.0	0.0	0:00.00	agetty
217	root	20	0	107016	23232	13768	S	0.0	0.3	0:00.07	unattended-upgr
260	root	20	0	17284	7540	6660	S	0.0	0.1	0:00.02	systemd-timedat
299	root	20	0	3188	1108	980	S	0.0	0.0	0:00.00	SessionLeader
300	root	20	0	3204	1248	1108	S	0.0	0.0	0:00.00	Relay(301)
301	ajithmu+	20	0	6076	5376	3656	S	0.0	0.1	0:00.03	bash
302	root	20	0	6704	4672	3884	S	0.0	0.1	0:00.00	login
345	ajithmu+	20	0	20308	11476	9352	S	0.0	0.1	0:00.07	systemd
346	ajithmu+	20	0	21156	3596	1872	S	0.0	0.0	0:00.00	(sd-pam)
370	ajithmu+	20	0	6064	5336	3668	S	0.0	0.1	0:00.00	bash
444	ajithmu+	20	0	9416	5788	3548	R	0.0	0.1	0:00.01	top
528	root	20	0	25408	3892	2224	S	0.0	0.0	0:00.00	(udev-worker)
529	root	20	0	25408	3892	2224	S	0.0	0.0	0:00.00	(udev-worker)